Deep Learning Based GPS Guided Drone for Tomato Crop Disease Detection in Precision Agriculture

KCT SWAMY
Electronics and Communication Department
G.Pullaiah College Of Engineering and Technology(JNTUA)
Kurnool, India
kctswamy@gmail.com

GUNI REDDY CHARAN KUMAR REDDY
Electronics and Communication Department
G.Pullaiah College Of Engineering and Technology(JNTUA)
Kurnool, India
charanreddy2908@gmail.com

KADUMURUGANDLA SRI GURU DATTA
Electronics and Communication Department
G.Pullaiah College Of Engineering and Technology(JNTUA)
Kurnool, India
srigurudatta368@gmail.com

BEKKASULA VINAY
Electronics and Communication Department
G.Pullaiah College Of Engineering and Technology(JNTUA)
Kurnool, India
vinayrbekkasula@gmail.com

***Abstract*—Agriculture plays a vital role in the economy, but crop diseases significantly reduce productivity and quality. Among various crops, tomato plants are easily affected by many types of diseases that can spread fast and cause big losses if not found early. Traditional disease monitoring methods rely on manual inspection and uniform spraying of pesticides, which are time consuming, labour intensive, and often lead to excessive chemical usage. In this project we focus on tomato crop disease detection using an autonomous drone . The drone is integrated with a Pixhawk flight controller and GPS module, which enabling to navigate the drone autonomously with predefined flight path created using Mission Planner software. A Raspberry Pi with the camera takes clear images of the tomato plants while the drone is flying on the field. The images taken by the drone are analyzed on the Raspberry Pi using a ResNet deep learning model that has been trained to recognize different tomato diseases, which achieved 99% accuracy. When the system detect an infected plant, it classify the type of disease, suggests the right chemical treatment, and saves the exact GPS location of the affected plant. All important information, such as the type of disease, its location, and the suggested treatment, is saved for future study and helps for precision spraying.**

Keywords—Tomato disease detection, Autonomous drone, GPS locations, ResNet model, Precision agriculture.

I. INTRODUCTION

Such a large and growing population makes it very difficult for the country to produce enough food for all of its citizens. Agriculture is quite vital to the economy of the nation and sustains millions of livelihoods. India produces about 20.7 million tonnes of tomatoes in a year ranking second in the world next only to China. The major tomato producing states are Andhra Pradesh, Madhya Pradesh, Karnataka, Gujarat and West Bengal, accounting for the majority share of the country tomato production. In the Kurnool district of Andhra Pradesh, A total extent of 7,900 hectares is covered by tomatoes. Although India is one of the leading producers of tomatoes farmers face many difficulties in getting better yield and quality. Things like vulnerability to plant diseases, fluctuating market prices and unpredictable weather further deteriorate productivity at work. Diseases can spread in no time and are costly thus, They hurt the economy's food supply. Most of the farmers still rely on the conventional method of disease detection in crops manually, and spreading chemicals in fields is also uniformly done. These methods are highly time consuming and energy consuming and are very inefficient and hazardous. Farmers face the risk of falling ill due to excessive use of fertilizers and pesticides. Pesticides can lead to headaches, fever, vomiting [2].

The way farmers in India are doing in 2025 shows how hard they work and how hard it really is for them. Even with new technologies, many farmers still have problems with bad weather and bad soil, not enough water and prices that go up and down in the market. All of these together affect how much money they make and the amount of work they do. Most farmers, especially small ones, have to deal with small plots of land which are far apart, not many machines and obtaining inexpensive loans and insurance. Also, the rather vintage methods of checking the crops by hand and spraying them with chemicals all over is not healthy for them the environment and the amount of crops grown. Using new technologies and help from the government are changing the way of farming. in India. [3] Now, farmers will be able to use drones, AI and mobile apps that will keep them updated on Give them useful information. These new ideas help them use their resources better, waste less and find better ways to sell their goods[4]. One step at a time, these changes are making Indian farming stronger, more sustainable, and ready for the future.

This work focuses on the development of an autonomous drone for assisting farmers in easy Detection of diseases in tomato crops, the drone is fitted with a Pixhawk flight controller with GPS that lets it automatically follow planned flight paths. A Raspberry Pi with a high resolution camera modules take detailed pictures of tomato plants while running. These images are Analysed using AI algorithms based on ResNet architecture, it identifies and classifies diseases, tag the affected regions with GPS coordinates, and store the related data. With this In an integrated set up, one is able to monitor the field at all times for accurate disease finding and correctly map out infected areas. In turn, it will help farmers to grow more crops, use fewer Reduce the use of harmful chemicals and manage better based on data.

II. LITERATURE REVIEW

Paper [5] proposed a system that adopts deep learning and IoT technologies to improve track the health of tomato plants. This research trains a convolutional neural network (CNN) to

identify different diseases of tomato leaves using real-time images acquired through a camera. The system will also monitor some key conditions within the farm, such as temperature, humidity, and moisture levels in the soil. The system monitors these continuously and also checks for important farm conditions such as temperature, humidity, and soil moisture. Sensors connected to IoT boards like NodeMCU, which collect these readings and forward the data on to platforms like Blynk for real-time monitoring and fast alerts. The setup not only automatically finds the plant, diseases, but it also helps in precision farming whereby irrigation is turned on or off depending on how wet the soil is. It gives farmers an excellent means to care better for their crops, because. It combines AI-based disease detection, environmental sensing, and mobile notifications. However, some challenges still remain when using the system in real-world conditions, especially in maintaining high accuracy and identifying the exact spots of the affected plants. A drone based system [6] is developed that detect diseases in crops using convolutional neural CNNs. In practice, the drone flies above plants and captures high resolution images that later processed and analysed to look for signs of infection. The model is much more accurate than traditional detection methodologies by integrating pre-processing, feature extraction, and classification stages. There is also an Android app that shows real time maps of where diseases found, and advice given to farmers. This speeds up the process, it will also make monitoring crops more efficient to help in disease control and sustainable farming.A self-manned flying hexacopter drone has been made for the identification of diseases in plant leaves. [7] The system has fitted a Raspberry Pi with a camera that clicks high-resolution pictures of the crops. The system clicks pictures, followed by cleaning by the removal of noise. Further, it segregates the areas of the leaves and then uses machine learning models such as CNN and SVM in order to study the textures of these images for the identification of any signs of disease. Currently, this system can identify the leaf diseases at an accuracy of about 71%. The drone it works in the field because it's light and can have a stable flight design based on basic principles of aerodynamics. This setup generally helps automate disease monitoring and reduces the need for any manual inspection to a great extent. However, more work is required to be done to make the entire system more accurate and practically applicable on a large scale. We propose a lightweight model for disease detection in tomato leaves based on the modified YOLOv8 configuration with adaptive kernel convolution, which can modify the shape of the kernel. This helps the network to find different disease patterns more precisely than the normal convolution because in regular convolutional layers, they use a fixed kernel shape. To make it faster and quicker, we added modules like C2f AKConv and GSConv to the model. These modules reduce computational overhead. Then, we applied the model to a large dataset of tomato leaves with some extra data and achieved a mAP50 of 94.9%. Overall, it is pretty fast and resource efficient. It works well in real time, making it suitable for use in drones and precision farming.

The work [9] presents a system for multiple disease detection on tomato leaves. By using a modified YOLOv7 model, one can have the model's attention mechanisms enabled by techniques such as SimAM and DAiAM to focus on the salient parts even when the field conditions are not good. Researchers also used image preprocessing, methods like SIFT, for finding the disease spots more accurately. They tried the model on eight different diseases, namely bacterial spot, early blight, and mosaic virus and found it was right about 98.8% of the time. The system combines deep learning with some traditional image processing to help farmers deal with Tomato diseases better in real life.

III. METHODOLOGY

Mission planner software tells the automated drone with the camera where it should go , which helps it cover the whole field in a structured way. The camera shows clear pictures of the tomato plants at regular intervals when the drone starts flying. A Raspberry Pi on the when the drone gets these images combined with a pretrained ResNet model, they are going to check the items in real time for any sign of disease, type of disease and exact GPS location recordings in the system are automated from the drone when it finds a sick plant. This will help the farmers by locating the exact areas that need work, saving all these data for later analysis. This method utilizes autonomous navigation, real stime AI enabled analysis and spatial. This also allows recording, reducing manual work and increasing efficiency, with very high precision field disease management.

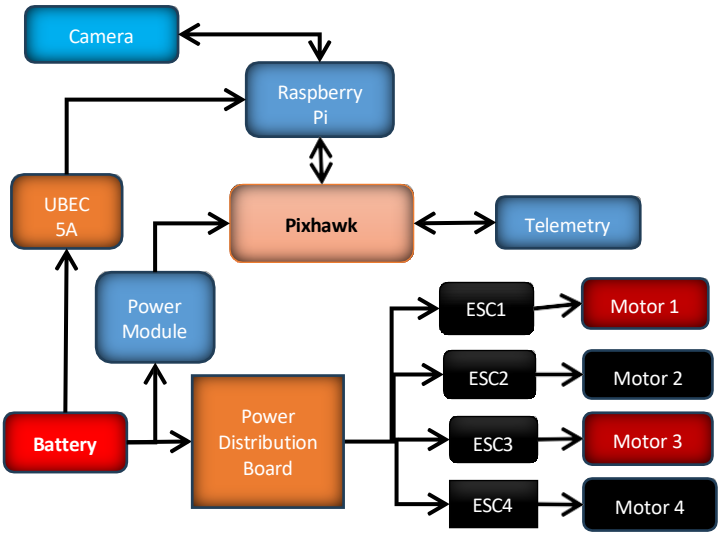


Fig 1: Block diagram of the drone

Pixhawk (APM 2.8): In a drone, the APM 2.8 flight controller is used. The Arduino Mega 2560 microcontroller makes it, and it has important sensors like a gyroscope, accelerometer, magnetometer, and barometer. These sensors let the controller know where the vehicle is, which way it is facing, and how it is moving in real time. The compass that comes with the controller makes it easier to find your way by letting it know which way it is facing. It is linked to GPS so that the car can fly on its own by following GPS waypoints that have already been set. Mission Planner software lets users set up and control the APM 2.8. They can program flight paths and watch the vehicle in real time. Electronic Speed Controllers send speed commands from the flight controller to the motors to keep the vehicle stable in the air. PPM and SBUS are two types of radio receivers that connect to the controller to get user commands. Each type has its own way of communicating and level of reliability.

GPS Module: The GPS module sends location data to the drone's flight controller APM 2.8 through wires that connect

the two. Most of the time, the GPS module connects through serial communication protocols such as UART. It makes use of pins for power, ground, and data transfer. Once connected, the GPS module gets signals from satellites that help the drone figure out where it is on Earth. This location information is then sent to a flight controller, which lets the drone fly on its own along paths that were previously set, stay in one place, or go back to where it took off. In modern systems, it is of great importance since it is basically the basis of everything from waypoint navigation altitude holding, navigation, and landing safely.

Reference	Methodology	Key Findings	Contributions	Research Objective	Key Challenges
[10]	Used CNN with VGG-16 and AlexNet models on tomato leaf images.	Achieved 91% accuracy in detecting 6 tomato leaf diseases.	Small, fast, and accurate model suitable for farmers.	To detect tomato leaf diseases early and correctly	Imbalanced dataset classes
[11]	Light CNN with 10 layers for early/late blight detection	99.7% accuracy, real-time drone deployment	Simple model working well in real-time	Accurate, fast blight disease detection.	Focussed on only 2 diseases
[12]	Deep Learning, ML	Accurate tomato disease detection	New multi-step detection method	Tomato disease detection with improved segmentation and classification	Dataset complexity, environmental variability
[13]	IoT sensors, Federated Learning	Farm sensor data helps monitor crops	Novel HFL framework for scalable real-time tomato crop monitoring	Easier crop monitoring on many farms	Network heterogeneity, distributed data privacy
[14]	CNN with generative adversarial networks, data augmentation	High performance tomato leaf disease classification (>99% accuracy)	Improved training and testing	High accuracy disease classification on tomato leaves	Data imbalance, generalization to field conditions
[15]	CNN,transfer learning	Effective tomato crop disease classification (88% test accuracy)	New CNN architecture optimized for tomato leaf disease	Build efficient CNN for tomato disease recognition	limited dataset size
[16]	CNN with two convolution and pooling layers	Competitive tomato disease detection with acceptable accuracy	Simple CNN model outperforming some pre-trained networks	Disease detection on tomato crops using deep CNN	Dataset limitations, model optimization
[17]	Drone with NDVI and semantic segmentation, SVM classification for pesticide spraying	Dual-index health monitoring, autonomous spraying	Holistic drone platform integrating edge computing for farm management	Automate crop health monitoring and targeted pesticide application	Computational complexity, robust segmentation
[18]	YOLOv8 object detection on drone image classification	Real-time wheat disease detection with good precision and recall	Drone-powered real-time disease localization using YOLOv8	Develop efficient wheat disease detector	Visual similarity of diseases, small dataset
[19]	AI, CNNs on drone imagery, TensorFlow/OpenCV processing	Early crop disease detection, precise automated monitoring	Integrated AI-drone system enabling real-time disease diagnosis	Applying drone and AI integration for precision agriculture	Computational resources, dataset diversity, deployment

Raspberry Pi: A Raspberry Pi used in a drone works like a small computer that helps the drone operate more intelligently and independently. It connects to the flight controller, such as an APM 2.8 or Pixhawk, using a USB or UART cable. In this setup, communication by Raspberry Pi is through MAV Link is a protocol that exchanges important data such as position, speed, and sensor readings. The flight controller manages the basic flying functions, the Raspberry Pi takes care of more advanced tasks such as controlling the camera, processing images, detecting objects, or planning autonomous routes. It will connect to the internet or to a ground station via Wi-Fi, which allows easy monitoring and control from a laptop. It is connected by a 5V power supply from the drone battery. The computing power of Raspberry Pi makes it ideal for research projects, smart drone systems, and real time video or sensor based applications.

ESCs: Electronic Speed Controllers act as the messengers between the drone's battery, motors, and flight controller. ESCs regulate the speed of the brushless motors in the drone. When we need to change the position of the drone, the speed of the motors should be regulated. ESCs take the battery's direct current and turn it into the three-phase alternating current that powers the drone's brushless motors. They use something called Pulse Width Modulation (PWM)—think of it as a super-precise dimmer switch to control how fast or slow each motor spins. Without ESCs the motors run at one speed, like a car stuck in one gear. Modern ESCs are used for handling high power without overheating, more efficient flights.

Brushless Motors: The most common motors used in drones are brushless motors, which, when compared to other motors, last longer and work better. Two of the main parts are a stator containing copper coils that remain fixed and a rotor with magnets that rotate. When electricity passes through the coils, a magnetic field is created that helps the motor to rotate. The ESCs maintain the speed and direction of the motor by sending electrical signals. These motors are quiet, smooth, and energy-efficient thus, flying is easy for drones. The strength of a motor is sometimes represented by its KV rating unit which tells the speed it spins per volt. Generally, brushless motors play an important part in giving the drone power for stable and fast flights with minimal maintenance.

Camera: A drone camera is a small camera attached to a drone that captures photos and videos from above. It connects to the Raspberry Pi the camera sends live video to a ground station while flying and thereby the user can see what the drone sees. Many cameras on drones are mounted on a gimbal, which keeps the camera steady and prevents shaky footage when the drone moves. There are different types of drone cameras for different uses. Normal RGB Colour cameras take photos and video, while thermal cameras detect heat for rescue or inspection tasks, and multispectral cameras, which are utilized in agriculture for plant and soil monitoring. Some drones come with inbuilt cameras, while others make use of detachable cameras, such as Go Pros. It is helpful in making movies, farming, mapping, and even they keep an eye on things, taking clear, steady, high quality pictures and videos from the sky.

Telemetry: Drones use a 433 MHz telemetry module to transfer the data between drone and the ground station. This frequency band is known for offering reliable and stable

communication over long distances. One module connects to the drone's flight controller, while the other connects to a computer or laptop. This wireless link provides the receiving ground station with real-time information the drone transmits about its position regarding GPS, altitude, speed, and battery conditions. The telemetry module makes use of the MAVLink protocol, which basically allows the user to monitor parameters remotely in software like Mission Planner or QGroundControl. It also allows users to send commands, upload flight paths, and even allow for running the drone remotely if need be. The range on this 433MHz telemetry kit typically lies within the range of 3 to 5 kilometres, but it depends on the quality of the antenna and surroundings. It uses frequency hopping and error correction to minimize interference in the signal and hold a steady connection. This is lightweight and compact, making it compatible with Pixhawk flight controllers and therefore good for research, mapping, and other autonomous drone projects.

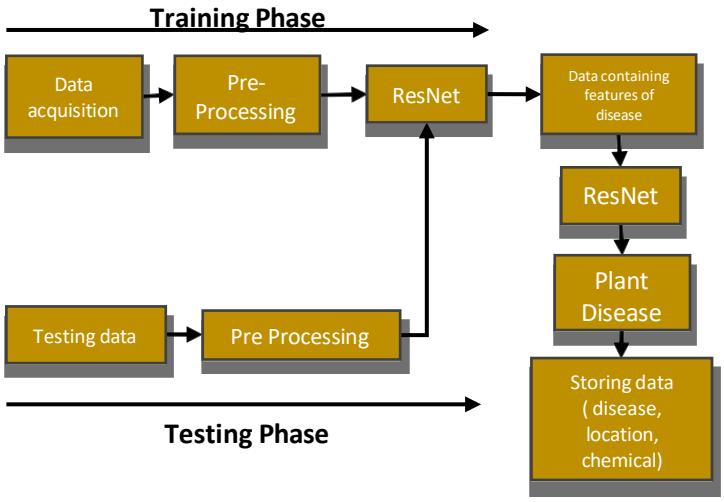


Fig 2: Block diagram of the proposed model

Training Phase:

Data Acquisition: For model training Fig 2, a publicly available tomato leaf disease dataset was utilized instead of images captured from the field. The dataset includes plenty of clear, high-quality images of tomato leaves affected by different diseases such as Bacterial spot, Early blight, Late blight, Leaf Mold, Septoria leaf spot, Spider mites Two-spotted spider mite, Target Spot, Yellow Leaf Curl Virus, mosaic virus as well as healthy leaves. Since these images come from different environments and growth stages, the model becomes better at recognizing diseases under varied conditions. Public datasets like this are commonly used in research because they make experiments easier to repeat and allow results to be compared with other studies.

Pre-Processing: Each was resized to the same dimensions and adjusted so that the pixel values fell within a similar range, then simple filters were applied on top to remove noise and enhance finer details in the pictures. We also added changes to rotate, flip, and change colors to give the model even more things it could learn from. This not only gave the model more number of training samples but improved its general performance regarding illumination and leaf conditions.

ResNet:

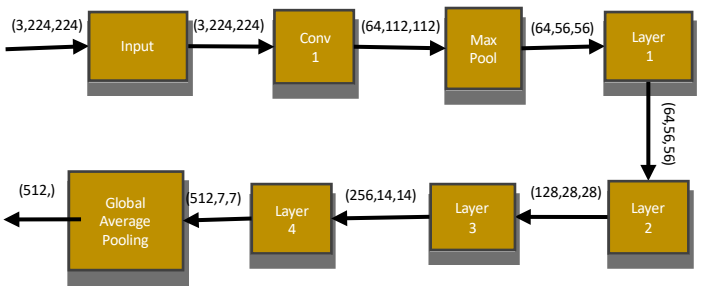


Fig 3: ResNet

Figure 3 presents a model designed for the task of classifying tomato leaf diseases accepting input in the form of RGB images with a 224×224 pixels size. This picture enters the first convolution layer, which is responsible for extracting basic features such as edges and textures through 64 filters this means the output of this layer is 64×112×112. Further this is reduced to 64×56×56 through max pooling. This saves the model from useless processing by focusing on the most informative parts of the image. The first residual blocks are known as Layer1 in the next step. They maintain the output in the size of 64×56×56. These blocks are used as shortcuts so that the model may learn complex patterns without forgetting what it learned from the previous steps. Then comes Layer2 where the channel size is increased to 128 and the spatial size is reduced to 28×28. So, this layer may catch more detailed patterns in leaves. Next Layer3 produces 256×14×14 features and captures deeper patterns hence useful for disease symptoms. The last residual blocks or Layer4 produce 512×7×7 features. They are high-order features the classifier will use to make the final prediction. Then comes the Global Average Pooling step, compressing each 7×7 feature map into a single number. In the end this returns a 512-dimensional vector containing the key details from all over the image.

Classifier: A classifier is basically the part of the model that makes the final call about what's in an image. After ResNet pulls out the important features from a tomato leaf, the classifier which has a few fully connected layers with a softmax at the end looks at those features and decides which disease the leaf could have.

Testing Phase:

Input Acquisition: The system takes new images of tomato leaves during testing. These can be taken from drone camera flies in the field.

Preprocessing: These steps involve resizing, normalizing and noise reduction, just as in training, to make the model work on the images taken by the drone and avoid errors due to inconsistent data.

Feature Extraction and Classification: The preprocessed images are sent to the trained ResNet model. It automatically picks out the most important features, and then the classifier decides if the leaf is healthy or shows any evidence of a specific disease.

Result Logging and Storage: For each image, it records the predicted disease, the confidence of the prediction, extra information like suggested chemical, product, GPS location from the drone. The data is then saved for later analysis and helps in planning interventions in the field.

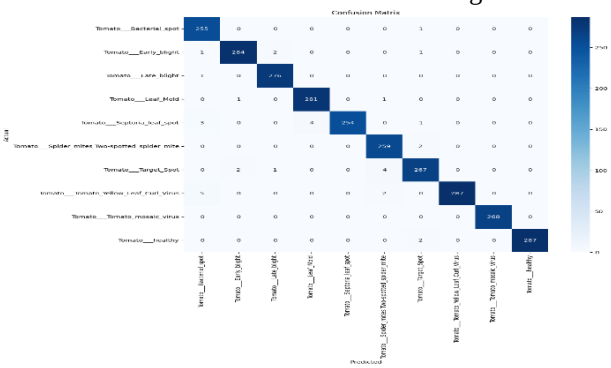
IV. RESULT

Fig 4: Autonomous quadcopter drone

Figure 4 is an autonomous quadcopter drone, employed to gather data in the field. It has flight control and image analysis systems installed to assist it in moving accurately and detecting plant diseases on the ground. In this paper, the ResNet-18 model was trained and tested for different kinds of tomato leaf diseases using a well organized set of images. There were 18,345 images in this dataset which were divided into 10 groups. Some of them were for healthy leaves, and some of them were for common diseases of tomatoes. We used PyTorch on a CUDA-enabled device to accelerate training, and used a fixed random seed (42) for all experiments so that the results would always be the same time. We used stratified sampling so as to keep the class proportions balanced, splitting the dataset into three parts 70% for training, 15% for validation, and 15% for testing. The model was making small changes to the images, such as cropping, flipping, rotating or changing their colours. This helped the model learn how to handle different situations. For the validation and testing stages, we only resized and normalized the images to ensure that the evaluation remained fair and consistent.

Fig. 5: A matrix of predictions for the tomato leaf disease

A confusion matrix is an informative diagram that can



visually display how well a model for classification performs. It shows how a model predicted labels compare to the real labels. The real class is shown is the actual class for every row in the table and the predicted class is in each column. The values on the diagonal show the correct predictions while other values show where the model went wrong. High diagonal values in the confusion matrix of Figure 5 indicate that most tomato leaf diseases, as well as healthy samples are correctly identified by the ResNet-18 model. Small numbers elements outside the diagonal show only a few misclassifications, which is enough to establish that the model can accurately tell the classes apart and can be accurate in detecting diseases in tomato plants. The confusion matrix gives a clearer view of the model performance in detail while presenting both the strengths and weaknesses, So it can provide a better understanding than general measures like accuracy or precision.

Classification Report:

	precision	recall	f1-score	support
Tomato___Bacterial_spot	0.96	1.00	0.98	256
Tomato___Early_blight	0.99	0.99	0.99	288
Tomato___Late_blight	0.99	1.00	0.99	277
Tomato___Leaf_Mold	0.99	0.99	0.99	283
Tomato___Septoria_leaf_spot	1.00	0.97	0.98	262
Tomato___Spider_mites Two-spotted_spider_mite	0.97	0.99	0.98	261
Tomato___Target_Spot	0.97	0.97	0.97	274
Tomato___Tomato_Yellow_Leaf_Curl_Virus	1.00	0.98	0.99	294
Tomato___Tomato_mosaic_virus	1.00	1.00	1.00	268
Tomato___healthy	1.00	0.99	1.00	289
accuracy			0.99	2752
macro avg	0.99	0.99	0.99	2752
weighted avg	0.99	0.99	0.99	2752

Table 1 presents the performance of the ResNet-18 model on detecting a variety of diseases affecting tomatoes, as well as healthy leaves. It yielded very high results in all categories, showing an excellent balance between precision and recall. The macro average, which considers all classes equally, and the weighted average, considering class size, reached up to 0.99 for both. This means that system predictions were almost always correct, whether it was looking at a sick leaf or a healthy one. As this table indicates, this model is reliable and confident in finding even small differences between diseases. This can prove that this model can be helpful for real-world monitoring and management of crops.

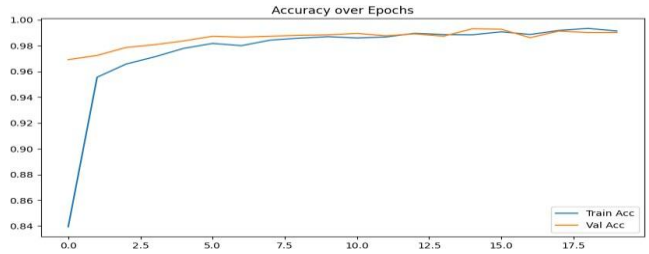


Fig 6: Accuracy of the model

Figure 6 shows the variation in the model's accuracy during training. The accuracy was low at First, which is normal because the model was still learning. As training went on, the accuracy of both training and validation rose quickly and then levelled off at about 99%. It means that the model generalizes well for new, unseen data since both lines are fairly close together with only small gaps between them. This means that the model design, along with careful data preparation, helped it learn quickly and make accurate predictions.

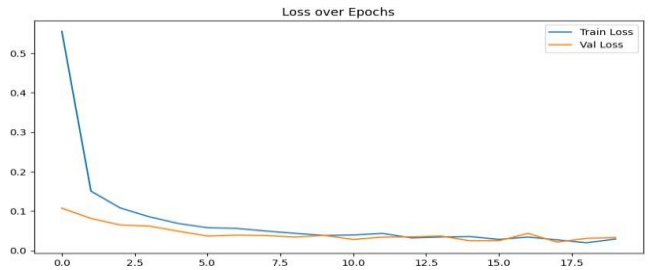
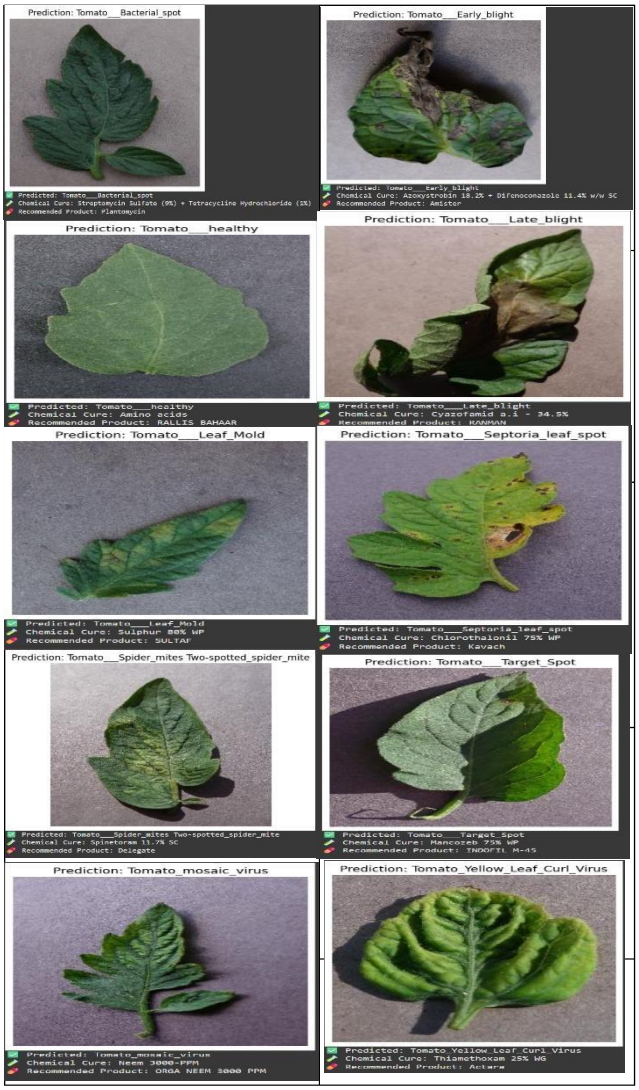


Fig 7: Loss of the model

In Figure 7 the model loss over the same period. During the first phase, there was high loss of the model, as one could expect since the model was yet to learn anything. With continued training, the loss continued to drop at a high rate, indicating the model was acquiring informative patterns from the images. Over time, the curves for both training and validation came together and stayed low, showing that the model was learning properly without just memorizing the training data.



V. CONCLUSION

The project depicts that the drone can make the identification and management of disease in tomato plants easier. Based on its flight over the field and information gathering, a drone can very quickly identify non healthy plants and location, type of disease, and appropriate chemical treatment. This will save time, decrease manual work, and chemicals in a more appropriate manner. In the future, it can be developed for other crops too. Including features like automatic spraying and organised data storage that help farmers to take good care of their fields. Generally, this technology can help crops to stay healthy and give more yield.

VI. FUTURE SCOPE

This information can be useful in the future when the system automatically decides how much of the chemical is actually needed for every disease and loads it into different containers of the drone spraying mechanism according to the kind of disease found. This means that the drone can use only the right amount of the right chemical in areas that are infested, hence reducing waste and ensuring that the diseases are treated properly. The system can be improved even further to deal with other crops such as chili, cotton, brinjal and maize. It will make the proposed system useful in a wide range of agricultural applications such as finding diseases and their spraying at the right time.

REFERENCES

[1] A. AKSOY and H. C¸ . KAYMAK, "Tomato production quantity estimates for 2023-2027 with arima model: Evidence from leading producing countries including turkey." Scientific Papers Series Management, Economic Engineering in Agriculture & Rural Development, vol. 24, no. 3, 2024.

[2] W. Zhou, M. Li, and V. Achal, "A comprehensive review on environmental and human health impacts of chemical pesticide usage," Emerging Contaminants, vol. 11, no. 1, p. 100410, 2025.

[3] R. Singh and S. Singh, "A review of indian-based drones in the agriculture sector: Issues, challenges, and solutions," Sensors, vol. 25, no. 15, p. 4876, 2025

[4] A. Nag, P. R. Chanda, and S. Nandi, "Mobile app-based tomato disease identification with fine-tuned convolutional neural networks," Computers and Electrical Engineering, vol. 112, p.108995, 2023.

[5] A. H. Deepak, A. Gupta, M. Choudhary, and S. Meghana, "Disease detection in tomato plants and remote monitoring of agricultural parameters," in 2019 11th International Conference on Advanced Computing (ICoAC). IEEE, 2019, pp. 28–33..

[6] R. Sagar, B. Arthi, P. S. Prasad, R. S. Prajwal, and V. Sanjana, "Drone based crop disease detection using ml," in 2024 Third International Conference on Distributed Computing and Electrical Circuits and Electronics (ICDCECE). IEEE, 2024, pp. 1–8.

[7] L. N. Thalluri, S. D. Adapa, A. V. N. Sarma, S. N. Venkat et al.,"Drone technology enabled leaf disease detection and analysis system for agriculture applications," in 2021 2nd International Conference on Smart Electronics and Communication (ICOSEC).IEEE, 2021, pp. 1079–1085.

[8] B. Ji, H. Li, X. Jin, J. Zhang, F. Tao, P. Li, J. Wang, and H. Fan, "Lightweight tomato leaf intelligent disease detection model based on adaptive kernel convolution and feature fusion," IEEE Transactions on AgriFood Electronics, 2024.

[9] "M. Umar, S. Altaf, S. Ahmad, H. Mahmoud, A. S. N. Mohamed, and R. Ayub, "Precision agriculture through deep learning: Tomato plant multiple diseases recognition with cnn and improved yolov7,"IEEE Access, vol. 12, pp. 49 167–49 183, 2024.

[10] D. Shanthi, K. Vinutha, N. Ashwini et al., "Tomato leaf disease detection using cnn," Procedia Computer Science, vol. 235, pp. 2975–2984, 2024.

[11] S. U. Islam, G. Schirinzi, and S. Maqsood, "A light weight cnn based architecture for the detection of early and late blight disease in tomato plants in real-time environment," in IGARSS 2024-2024 IEEE International Geoscience and Remote Sensing Symposium. IEEE, 2024, pp. 2819–2822.

[12] R. Bora, D. Parasar, and S. Charhate, "A detection of tomato plant diseases using deep learning mndlnn classifier," Signal, Image and Video Processing, vol. 17, no. 7, pp. 3255–3263, 2023.

[13] H. Devaraj, S. Sohail, M. Ooi, B. Li, N. Hudson, M. Baughman, K. Chard, R. Chard, E. Casella, I. Foster et al., "Ruralai in tomato farming: Integrated sensor system, distributed computing, and hierarchical federated learning for crop health monitoring," IEEE Sensors Letters, vol. 8, no. 5, pp. 1–4, 2024.

[14] A. Guerrero-Iba˜nez and A. Reyes-Mu˜noz, "Monitoring tomato leaf disease through convolutional neural networks," Electronics, vol. 12, no. 1, p. 229, 2023.

[15] S. K. Swarnkar, L. Dewangan, O. Dewangan, T. M. Prajapati,and F. Rabbi, "Ai-enabled crop health monitoring and nutrient management in smart agriculture," in 2023 6th international conference on contemporary computing and informatics (IC3I), vol. 6. IEEE, 2023, pp. 2679–2683.

[16] G. Sakkarvarthi, G. W. Sathianesan, V. S. Murugan, A. J. Reddy, P. Jayagopal, and M. Elsisi, "Detection and classification of tomato crop disease using convolutional neural network," Electronics, vol. 11, no. 21, p. 3618, 2022.

[17] A. Tendolkar, A. Choraria, M. M. Pai, S. Girisha, G. Dsouza, and K. Adithya, "Modified crop health monitoring and pesticide spraying system using ndvi and semantic segmentation: An agro-copter based approach," in 2021 IEEE International Conference on Autonomous Systems (ICAS). IEEE, 2021, pp. 1–5.

[18] K. Neha, A. Arya, and R. Singh, “Wheat disease detection using yolo and drone-captured images,” Journal of Graphic Era University, pp. 411–438, 2025.

[19] N. Naseer, S. Sahana, S. Bedre, S. Sangeetha, and S. Sneha, “Ai-based drone for crop disease detection in precision agriculture,”international Journal of Engineering Research & Technology(IJERT), vol. 13, no. 4, p. 268, 2024.